

C:\Users\mthob\OneDrive\Documents\NetBeansProjects\FederatedIDSArtefact\src\main\java\com\registration\app\federatedid

```
1
2
3
4
5 package com.registration.app.federatedidsartefact;
6
7 /**
8  *
9  * @author mthobi
10 * SIMPLE IDS MODEL - This is a lightweight linear classifier suitable for resource-constrained IoT gateways.
11 */
12
13
14 public class Model {
15
16     public double[] weights;
17     public double bias;
18
19     public Model(int numFeatures) {
20         weights = new double[numFeatures];
21         bias = 0.0;
22     }
23
24     public int predict(double[] x) {
25         double sum = bias;
26         for (int i = 0; i < weights.length; i++) {
27             sum += weights[i] * x[i];
28         }
29         return sum >= 0 ? 1 : 0;
30     }
31
32     //model training method
33     public void train(double[][] X, int[] y, double lr, int epochs) {
34         for (int e = 0; e < epochs; e++) {
35             for (int i = 0; i < X.length; i++) {
36                 int pred = predict(X[i]);
37                 int error = y[i] - pred;
38
39                 for (int j = 0; j < weights.length; j++) {
40                     weights[j] += lr * error * X[i][j];
41                 }
42                 bias += lr * error;
43             }
44         }
45     }
46 }
47
48
```